



**Verified Carbon
Standard**

IMPROVED COOKSTOVES FOR BURUNDI RESTAURANTS

Project Description prepared by AERA Group

on behalf of OBEN E.A.C SA

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Prepared By	<i>AERA Group</i>
Contact	<i>28 Cours Albert 1er, 75008 Paris, France</i> <i>Ph +33 1 42 18 02 02 Fax +33 1 43 25 80 06</i> <i>contact@aera-group.fr, www.aera-group.fr</i>

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1 PROJECT DETAILS

1.1 Summary Description of the Project

OBEN E.A.C (SA) developed an improved cookstoves project for restaurants¹ of Burundi. The proposed small scale VCS project activity aims at:

1. Distributing institutional improved cookstoves (ICS) in restaurants of Burundi to replace currently used traditional open fire three-stone system (and traditional stoves); and
2. Switching from non-renewable wood fuel to a sustainable energy supply: briquettes made of renewable biomass waste. Compared to the currently used three-stone fires or traditional stoves, the advanced technology of ICS allows quicker heating-up, longer cooking and heat retaining with less wood-fuel as well as lower combustion fumes. It results in saving wood-fuel and associated expenses.

Along with the diffusion of such a stove to replace currently inefficient cooking systems, a renewable biomass supply-chain was set up, by sourcing unutilized biomass residues to produce renewable biomass briquettes and market it to the participating restaurants in replacement of their non-renewable wood fuel. OBEN will ensure a competitive and attractive price for using briquettes together with improved cook stoves to give incentives to the restaurants to switch 100% from the previous costly non-renewable wood fuel to the innovative briquettes made of renewable biomass wastes.² OBEN will ensure through a sale agreement and robust supply strategy that no shortage of briquettes will appear.

The distribution of up to six thousand ICS to restaurants supplied with renewable biomass briquettes, and the associated awareness and training campaigns will help halving these commercial entities' fuel use³ and turning it 100% renewable. Thus, the project will reduce greenhouse gas emissions by reducing the use of non-renewable biomass within the country, therefore slowing down deforestation.

¹ One restaurant should be understood as any commercial entities preparing meals. Bars and Hostels are covered under the proposed project and assimilated as restaurants.

² Thanks to ICS cook stoves, OBEN expect to replace 1 stère of wood by in average 20 kg of briquettes (based on experiences done in police camp (an average of 22 kg of briquettes per stère) and schools (an average of 18 kg of briquettes per stère) depending on Lower Heating Value of raw material used for the briquettes (indicative information is 3,300 kcal/kg briquettes for briquettes made of sugar mill wastes against 4,800 kcal/kg briquettes for briquettes made of pine needles). The report of test has been provided to DOE. Actually, one stère of wood is ~~val~~ at about 15,000 FBU and one bag of charcoal of 23 kg at 30,000 FBU whereas OBEN E.A.C SA plans to sell 1 kg of briquettes at 378 FBU (indicative price); satisfying the same thermal energy need, will thus cost from two to ten times less than in baseline situation according to one-site visit interviews.

³ Experiments to estimate the approximate renewable briquettes consumption per cookstove have been conducted and recorded, evidencing savings up to 90% in quantity and 50% time. Communities will save fuel and time

The expected annual amount of greenhouse gas reductions thanks to the project activity averages 131,082 tCO₂ eq and the total amount of greenhouse gas emissions reductions for the chosen crediting period is 917,572 tCO₂ eq. The type of GHG reduced is CO₂ emissions from substitution of fossil fuel consumption by similar consumers, equivalent to the non-renewable woody biomass saved by the project activity.

Baseline scenario

According to the World Bank, 172,000 ha (7% of land area) are forested in Burundi in 2010. In total, between 1990 and 2010, this country lost 40.5% of its forest cover or around 117,000 ha before getting back to 1990 level since 2015 (World.DataBank, 2020). In the current situation, wood products represent over 96.6% of the energy balance of Burundi (CIRAD, 2017).

1.2 Sectoral Scope and Project Type

The technology to be implemented by the Project activity falls into Sectoral Scope 1: Energy Industries (renewable sources), as it concerns the introduction of improved cooking devices supplied with renewable biomass sources to provide thermal energy to the restaurants and avoid consumption of non-renewable biomass. The Project activity concerns the provision of briquettes made of renewable biomass waste to restaurants for cooking activities together with the introduction of improved cooking devices supplied with renewable biomass briquettes to provide thermal energy to the restaurants, consequently avoiding consumption of non-renewable wood fuel. It falls into Type (I) project activities: renewable energy project activities with a maximum output capacity equivalent to up to 15 megawatts (or an appropriate equivalent). Moreover, the SSC-VCS project falls into Category (C) project activities: thermal energy for the user. The demonstration that the project is eligible as a small-scale activity is done in section B.2.

Project type	Project category	Output capacity of renewable equipment	Equivalent output capacity of renewable equipment
Type (I): Renewable energy projects	C. Thermal energy for the user	15 MW	45 MWth ⁴

1.3 Project Eligibility

- AMS-I.E category comprises activities to displace the use of non-renewable biomass by introducing renewable energy technologies, which is precisely the purpose of the project

⁴ Clean Development Mechanism Project Standard (Version 5.0), Section 8.2. project activity eligibility, §81. (a) &82. (c)

activity (the supply of renewable biomass-briquettes to ICS in restaurants); therefore category I.E is deemed adequate.

Indicator	Evidence
Survey results, national or local statistics, studies, maps, or other sources of information, such as remote-sensing data, that show that carbon stocks are depleting in the project area	In total, between 1990 and 2010, Burundi lost 40.5% of its forest cover or around 117,000 ha (World.DataBank, 2020).

2. It is also proven that non-renewable biomass has been used since 31 December 1989:

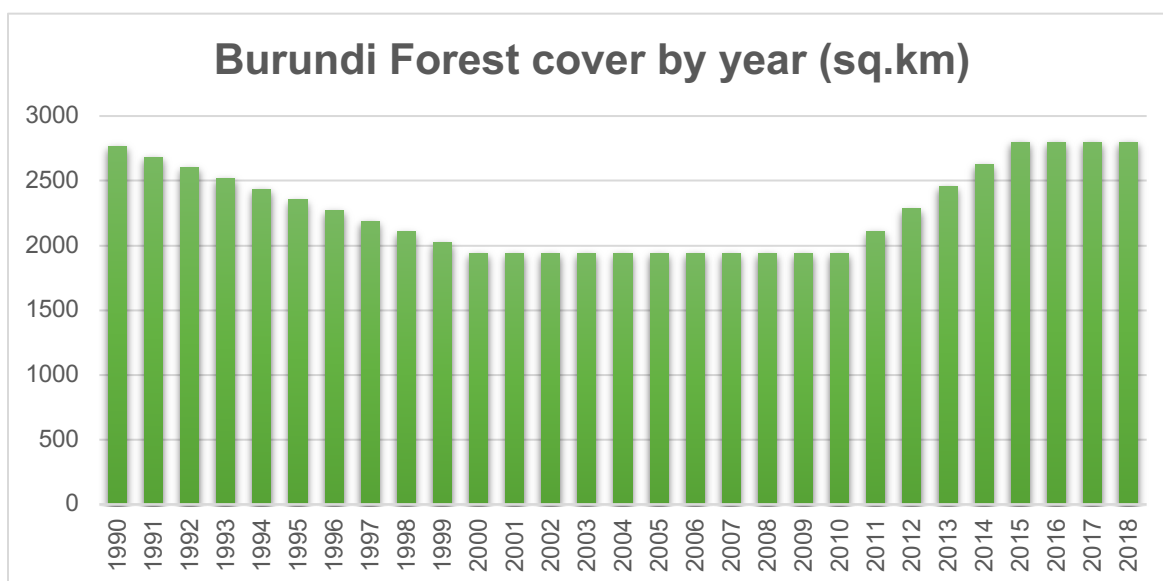


Figure 1: Change in forest cover in Burundi between 1990 and 2018

Moreover, it qualifies as a small-scale project activity as it will remain under the limits of small-scale project activity types during every year of the crediting period. Indeed, the appropriate capacity limit for Type I projects is 45 MWth, whereas each ICS thermal output is given at 4.82 kW according to certified tests⁵, hence a total effective output of 6,000⁶ appliances x 4.82 kWth = 28.920 MWth < 45 MWth.

⁵ Conservative assumption based on the maximum power output between the medium and large stove models: see test protocol and results.

⁶ According to project proponent experience and expectations, no more than six thousand ICSs should be distributed. However it is indicative information as the distribution between large and medium as well as the exact number of Institutional cookstove distributed may rise.

1.4 Project Design

The project has been designed to include a single installation of an activity.

Eligibility Criteria

This is not a grouped project therefore the section is not applicable

1.5 Project Proponent

Organization name	OBEN EAC (S.A.)
Contact person	Claver NDIZEYE
Title	Mr.
Address	B.P.4942, Bujumbura, Burundi
Telephone	0025776602600
Email	loicqueen@yahoo.fr

1.6 Other Entities Involved in the Project

Organization name	AERA Group
Role in the project	Carbon consultant and trader
Contact person	Alexandre Dunod
Title	Mr.
Address	28 Cours Albert 1er, 75008 Paris, France
Telephone	+33 1 42 18 02 02
Email	a.dunod@aera-group.fr

1.7 Ownership

As per VCS Program Definitions version 4, the project ownership is the legal right to control and operate the project activities. “Observatoire pour la Nature et l’Environnement” (also referred to as OBEN E.A.C. SA) is the project implementer and supplier of the improved cookstoves and renewable biomass, which retains the right of use of ER credits. The chain of ownership onwards to Aera Group is itself agreed upon by Emission Reduction Purchase Agreement.

1.8 Project Start Date

03/02/2020. The start date has been determined as the date of distribution of stoves and renewable briquettes starting on 03/02/2020.

1.9 Project Crediting Period

The project crediting period is 7 years renewable twice for a total of 21 years, starting from project start date specified above.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	x
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
03/02/2020 to 31/12/2020	119 112
01/01/2021 to 31/12/2021	159 692
01/01/2022 to 31/12/2022	159 692
01/01/2023 to 31/12/2023	159 692
01/01/2024 to 31/12/2024	159 692
01/01/2025 to 31/12/2025	159 692
01/01/2026 to 31/12/2026	159 692
01/01/2027 to 31/01/2027	13 307
Total estimated ERs	917,572
Total number of crediting years	7
Average annual ERs	131,082

1.11 Description of the Project Activity

In the baseline scenario (identical to the scenario existing prior to the implementation of the project activity), the prevailing technology in operation is the basic three stones cookstoves or traditional stoves broadly used in most Central African's urban settlements, fueled by non-renewable woodfuel.

Three stone cooking fires and traditional stoves are the cheapest methods of cooking. However, such practices also come with several issues:

- Fuel is wasted, as heat is allowed to escape into the open air.
- The inhalation of smoke causes health problems.
- The use of an open fire creates a risk of burns and scalds.

Under the current scenario, cookstoves are fueled by non-renewable biomass, i.e., batch deliveries of firewood and charcoal bags extracted from forest areas.

Technology to be implemented under the project activity

The list of the facilities, systems and equipment that will be installed and/or modified by the project activity includes:

- The distribution at an affordable price of up to six thousand Institutional Cookstoves to restaurants identified as participating customers to the proposed project.
- The setting up of a renewable briquettes supply chain from biomass residues.

Three (03) types of stoves are being distributed: a Turkish-manufactured robust design (Figure 2) available in medium (48 cm diameter) and large sizes (78 cm), and a locally manufactured Matawi stove. Compared to the currently used traditional stoves, the advanced design of the institutional improved cookstoves employed in the project allows quicker heating-up, longer cooking and heat retaining with less woodfuel as well as lower combustion fumes.



Figure 2: Improved Institutional cookstove with briquettes (left: Turc Medium, right: Matawi)
Improved cookstoves are designed by the Turkish company SOBAH and manufacturer specifications.

Table 1: SOBAH's and Matawi institutional improved cookstoves specifications

	Medium Turc ICS	Large Turc ICS	Matawi
Diameter	48 cm	78 cm	26 cm
Weight	46 kg (stove in box)	115 kg (stove in box)	5.1 kg
Cooking power output	2.49 kWth	4.82 kWth	1,45 kWth

Adoption	traditional cooking styles and posture	
Lifetime	5 years warranty	8 years
Age at the time of implementation	new	New

The project consists in switching 100% of the fuel consumption of the restaurants of Burundi from non-renewable biomass consumption for thermal applications into renewable sources. This will be possible thanks to the innovative introduction of briquettes made of renewable biomass wastes.



Figure 3: Briquettes made of renewable biomass wastes

Three production sites (Bujumbura, Rutana and Gitega) are expected to be set up for the supply of all restaurants of Burundi with briquetting machines Jumbo 90⁷. The current supply is ensured by local wood-fuel suppliers that will be included in supplying strategy of OBEN E.A.C SA.

How the technologies and know-how to be used are transferred to the host country

The technology and know-how being applied by the project is environmentally safe and sound since it will positively curb deforestation currently endangering natural forests by promoting sane and sustainable practices in the targeted areas of Burundi. It is made possible by highly efficient stoves and pioneer briquetting machines technology transfer to the Host party under the benefits of the Clean Development Mechanism. For the project, before renewable biomass fired ICS and renewable biomass distribution, a campaign of sensitization on the drawbacks and dangers to the ecosystem and populations using traditional cookstoves will be conducted. Besides distribution, after-sale services will comprise a suitable training, follow-up, and maintenance program for users to ensure that the renewable biomass fired ICS are properly and durably use.

1.12 Project Location

⁷ The details description and technical specification of briquetting machines are put in appendix 1 as they are not included within the project boundaries because each production site will supply independently the closest cities' restaurants and not any production site will be dedicated specifically to one city's restaurants. Therefore, emissions due to electricity consumption from the briquetting machines will be accounted as project emissions

Host Party(ies): The Republic of Burundi is the host country

Region/State/Province: etc. All provinces, starting by Bujumbura province.

City/Town/Community etc. All major urban centers, starting by Bujumbura.

Physical/Geographical location

The project is to take place in restaurants in Burundi urban centers, starting with Bujumbura city. As a reference, Bujumbura city center's geo-coordinates are 3° 22'34" S and 29° 21'36" E (cf -3.3761, 29.3600). The country-wide geographical area corresponds to the area where renewable biomass will be distributed and used in replacement of the former non-renewable woodfuel in ICS (Figure 4).



Figure 4: Area of implementation of the project activity.

The targeted restaurants are scattered throughout the country. The indicative number of participating restaurants⁸ per communes of Bujumbura city and second urban centers is given in Table below.

Table 2: Indicative Number of participating restaurants

BUJUMBURA MAIRIE			
COMMUNES	i	Population 2008 (pers)	Indicative Number of participating Restaurants
BUTERERE	1	28,371	75
BUYENZI	2	47,363	103
BWIZA	3	37,688	59
CIBITOE	4	50,899	55

⁸ Most of participating restaurants will be in Bujumbura city; however other restaurants from secondary urban centers such as Rumonge, Gitega, Ngozi, Kirundo, Kayanza etc. are targeted as well. All participating restaurants are not precisely identified at the time of project validation.

GIHOSHA	5	39,503	80
KAMENGE	6	50,070	89
KANYOSHA	7	59,181	97
KINAMA	8	49,776	45
KININDO	9	21,920	64
MUSAGA	10	43,735	18
NGAGARA	11	30,296	59
NYAKABIGA	12	20,883	35
ROHERO	13	17,481	34
Secondary urban centers	i	Population 2008 (pers)	Indicative Number of participating Restaurants
GITEGA	1	155,005	253
NGOZI	2	120,157	196
KIRUNDO	3	93,110	152
KAYANZA	4	97,252	159
RUMONGE	5	145,074	237

The project proponents have identified several renewable biomass resources⁹ throughout the country. The renewable biomass is from crop residues (like bagasse and coffee husks), forest litter (like pine needles) and timber residues (like sawmill residue).

OBEN E.A.C SA plans to operate three briquetting plants across the country, respectively in the vicinity of Bujumbura (West), Gitega (center) and Rutana (South-East), to adequately gather and process the surrounding biomass resources and supply the relevant urban centers' restaurants with minimal transportation. The first and second plant will operate in Bujumbura and Gitega respectively with supplies comprising coffee husks and pinus biomass residues, while the third one will operate in Rutana, supplied with sugar cane bagasse surplus from SOSUMO, the national sugar cane company.

1.13 Conditions Prior to Project Initiation

The baseline scenario is the same as the scenario existing prior to the implementation of the project activity please refer Section 2.4 (Baseline Scenario).

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

⁹ Renewable resource should be understood as per the definition provided in CDM-EB 23, Annex 18

There is no mandatory law or regulation in the host country requiring the introduction of ICS in Burundi, therefore the project is a voluntary effort by the project proponent. The project is following relevant laws, statutes, and other regulatory frameworks.

Therefore, the project does not violate any laws, statutes, or other regulatory frameworks in Burundi

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project neither has nor intended to generate any other form of GHG related environmental credit for the GHG emission reduction or removal claimed under the VCS programme. Although it was originally registered under the CDM by its former developers¹⁰, it was postponed and re-designed by its new developers, for certification and verification under the VCS only.

The project is currently being de-registered from the CDM.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has neither sought nor received another form of GHG-related environmental credits.

1.17 Additional Information Relevant to the Project

Leakage Management

Not applicable to this project activity. The project activity has been designed to include only new improve cookstoves. No used/old improved cookstoves will be transferred or deployed from outside the geographical project boundary to the project activity. Additionally, the project activity applies CDM Methodology: AMS-I.E. Version 12. According to the methodology, the default net to gross adjustment factor of 0.95 has been applied to account for leakage and therefore no surveys or no separate leakage management are required.

Commercially Sensitive Information

¹⁰ As per CDM Project 9911 : BQS improved cookstoves for Burundi restaurants, undergoing de-registration

No sensitive information has been excluded from the public version of the project description.

Sustainable Development

In addition to greenhouse gas emissions reductions, the project participants are confident that the proposed project activity will have the following significant impacts on Republic of Burundi's sustainable development.

- Socio-economical improvements

The project activity, by facilitating access to improved cooking technology, reduce restaurants' expenses from non-renewable biomass purchase, hence improving their value creation. Moreover, the project activity will foster employment in local communities throughout its lifecycle. The project will also provide direct employment opportunities for local people (improved cookstoves supply chain and distribution channels, renewable biomass supply chain and distribution channels, marketing campaign).

- Health benefits

Health hazards from combustion gases breathing and insufficient cooking time are likely to decrease. Respiratory diseases (infections, lung cancer, and asthma) due to carbon monoxide and other harmful particles related to incomplete and/or indoor solid fuels combustion rank among the worst threats to women and children's life in poor developing countries, along with malnutrition, HIV/AIDS and lack of clean water and adequate sanitation. By reducing indoor air pollution, ICS reduce the risk of respiratory diseases, especially for women and children. According to World Health Organization, 4 million of people die prematurely from illness attributable to household air pollution from inefficient cooking practices using polluting stoves paired with solid fuels. (World-Health-Organization, 2021).

- Environmental sustainability

The project will promote and stimulate awareness about renewable biomass benefits, the promotion of improved cooking stoves and forest conservation necessity, ahead of the Government's vision (EUEI, 2011).

- Technological benefits

The project will introduce better cooking technologies in restaurants, at the same time as developing the production of briquetting from renewable biomass. Demonstration and training programs will be carried out for ensuring good implementation of the project and transfer of technology. Therefore, the project complies with the national criteria for sustainable development.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

There was no harm identified from the project and hence no mitigations measures are applicable.

2.2 Local Stakeholder Consultation

Procedure followed to invite stakeholder comments

The national stakeholder consultation consisted of a public meeting with the identified stakeholders. Local and national policy makers and stakeholders were invited by post mail – see invitation material. Local population was invited through post mail to their representatives, through flyers diffusion and public displays.

Program of the public

Date of meeting: January 13th, 2012 at 9.00 am

Place: Olympafrica (hall of Olympic Stadium), Kanyosha Commune, Bujumbura Mairie Province

Language: Kirundi

Duration: 3 hours and 30 minutes

Meeting procedure:

- Introduction / Mayor of Bujumbura / DNA Permanent Secretary/ Project Developers
- Context and objectives of public consultation
- Project description
- Improved cookstoves
- Renewable biomass
- Presentation of the Clean Development Mechanism
- Environmental and socio-economic impacts
- Modalities of purchase and use
- Status of the project
- Video Spot Projection
- Questions and answers with participants

- Conclusion / Cocktail

Note: During this meeting, OBEN E.A.C SA jointly presented the project for restaurants and its other projects aimed at other Burundian communities (households, schools, polices, etc.).

Key invitees present were:

- Mr. Sef Sabushimike, Presidential advisor, Government of Burundi
- Ms. Bernadette Hakizimana, Director of Environment at the Ministry of Water, Environment, Land Management and Urban Planning (MEEATU), DNA Permanent Secretary
- Mr. Felix Ngengabanyikwa, Director of Forest Department at the Ministry of Water, Environment, Land Management and Urban Planning (MEEATU)
- Mr. Ezeckiel Nibigira, National Deputy, President of party CNDD-FDD's youth league
- Deogratias Suzuguye, Head of Penitentiary Matters, Ministry of Justice
- Serges Irambona, Head of National Police Logistics, Ministry of Public Security
- Mr. Niragira Félix, National Deputy (CNDD-DDD)
- Mr. Evrard Giswaswa, Bujumbura Mayor
- Anatole NIYONKURU, Secretary of Education Ministry

Media:

- Burundi National Radio and Television (RTNB)
- “Le Renouveau” newspaper

Figure 5: Picture of the stakeholders attending public consultation



Participant's questions/concerns	Project proponent's answers
Julien Ntarugera, Manager, Techno Solar Company. Is there a lifetime guarantee on your improved cookstoves? How does warranty work?	Yes, OBEN E.A.C SA's improved cookstoves will have a warranty of five years to the original owner against defects in materials and workmanship for the lifetime of improved cookstove. If a product ever fails due to a manufacturing defect, we will repair the product, without charge, or replace it, at our discretion. This warranty does not cover damage caused by accident, improper care, negligence, normal wear and tear, or the natural breakdown of colors and materials over extended time and use. Damage not covered under warranty will be repaired for a reasonable rate and return shipping will be charged
Have you considered local manufacturers, contributing to job creation, and lowering production cost? Why did you decide to outsource the manufacturing and assembling of improved cookstoves?	Local artisans and entrepreneurs cannot support a massive manufacturing and ensuring quality and traceability of improved cookstoves at such scale. Few years ago, the Office of the United Nations High Commissioner for Refugees (UNHCR) embarked on an initiative of improved cookstoves in Burundi. The scheme was unsuccessful for many reasons: the custom-built stoves were expensive and not appropriate for refugees cooking habits; dissemination was limited because stoves were made on-site at a slow rate by local artisans and entrepreneurs, and quality control was nonexistent. OBEN E.A.C SA has decided to outsource the manufacturing and assembling of improved cookstoves for quality and quantity reasons. OBEN E.A.C SA has a partnership with Sabah Enamel & Stove Industry base in Turkey to support the manufacturing and assembling of improved cookstove. The project leads to a technology transfer. The project will provide employment opportunities for local people to improved cookstoves supply chain and distribution channels, renewable biomass supply chain and distribution channels, campaign marketing.
Could the stove's high efficiency prematurely degrade the cooking pots?	No because wood & briquettes have a lower calorific value than woodfuel therefore such renewable fuels will be less harmful to the pots.

Participant's questions/concerns	Project proponent's answers
Ezeckiel Nibigira, National Deputy, Bujumbura Mairie and Rural Province [Full endorsement of the project, including congratulations to project developers, inputs on economic benefits to households and safeguard of existing forests] However, is it not possible to accelerate the project start? On behalf of a member of the "ADIDI" organization whose activity is nursery production, can you detail tree-planting operations that you intend to undertake during the project?¹¹	There are prerequisites to start the CDM project, including public consultation. CDM project needs to be registered by the UNFCCC, namely the CDM Executive Board. One of renewable biomass resources identified for the PoA "Renewable biomass fired improved cookstoves programme for households in Burundi by OBEN E.A.C SA" is biomass wood wastes collected from existing forests. The exploitation of this resource will be achieved by OBEN E.A.C SA under contract partnership with forests department. We are planning to plant 1,000 hectares per year for the duration of the project, starting from first carbon credits availability. The "Association ADIDI" was identified as a future partner to supply seeds of trees.¹²
Bernadette Hakizimana, Director of Environment at the Ministry of Water, Environment, Land Management and Urban Planning (MEEATU), DNA Permanent Secretary [Highlights on OBEN E.A.C SA project contribution to empowering women thanks to renewable biomass fired improved cookstoves achieving the following: Reduce the time and cost of procuring fuel, thereby freeing individuals for other productive activities. Reduce disease and save lives by decreasing exposure to indoor air pollution. Reduce the risk of violence against women a child gathering fuel in conflict areas.	

Participant's questions/concerns	Project proponent's answers
Venant Barindogo, IFDC [Supporting considerations on wood fuel consumption and the importance of raising awareness. Confirmation that their own experimentation with local manufacturing failed because of poor quality of the scrap metal and bricks used] Isn't the final cost still quite high compared to citizens' buying power?	The stove's cost must be aggregated with the significant monthly savings enabled on the fuel, hence a very profitable overall acquisition. [N.B. this answer was specifically given for the context of the parallel CDM initiative of OBEN E.A.C SA for households, although it remains valid in the case of communities which operating expenditures will also decrease].

¹¹ This question regarded the PoA "Renewable biomass fired improved cookstoves programme for households in Burundi by OBEN E.A.C SA" developed by OBEN E.A.C SA where renewable biomass will be produced thanks to sustainable management of existing Forests for households

¹² For the proposed project the biomass will come from agro-industrial wastes transformed into renewable briquettes.

Felix Ngengabanyikwa, Director of Forest Department at the Ministry of Water, Environment, Land Management and Urban Planning (MEEATU) Felix Ngengabanyikwa is suggesting that each improved cookstoves purchaser could plant a new tree at home to mitigate further the climate change effects. Do you accommodate for recycling of discarded improved cookstoves at the end of their use?	OBEN E.A.C SA has taken note of proposal and will analyze how to integrate this aspect in its programme. Anyway, its commitment to plant 1,000 ha per annum already exceeds this individual suggestion. Most parts of the improved cookstoves can be recycled. OBEN E.A.C SA will contact existing recycling industries that can collect and process such recyclables.
Jean Marie Masumbuko, Pastor of ECCA Kanyosha (Evangelical Church of Central Africa) How many improved cookstoves do you plan per households? ¹³ If the refractory brick breaks down, do you plan to replace them for free?	Each household might buy as many improved cookstoves as necessary if one ICS replaces one former traditional stove. Spare bricks will be available. Warranty application or replacement for a fee in case of misuse will be assessed on a case-by-case basis.

Participant's questions/concerns	Project proponent's answers
Salvator Mpangaje, Neighborhood Leader of Nyakabiga Could you tell me more about the product acquisition agreement between OBEN E.A.C SA and end user?	This agreement contains the terms and conditions that apply between OBEN E.A.C SA and improved cookstoves end user: A complete abandonment of the use of traditional stove(s) Adoption and sustained use of ICS. Exclusive use of renewable biomass with ICS Record of periodic consumption of renewable biomass Acceptance of periodical audits by monitoring agents. Such monitoring is needed for transparent verification in carbon projects and for improved dissemination with the highest adoption and the substitution of cooking practices.
Zacharie Tuyaga, Pastor of ECCA Ruziba (Evangelical Church of Central Africa) Is it forbidden that craftsman imitate and manufacture the improved cookstove?	No, it is not forbidden and could be included in later additions to the program. They are even encouraged to do the same, there is no restriction as the objective is to substitute as much renewable biomass or briquettes as possible instead of non-renewable woodfuel.

¹³ This question regarded the PoA "Renewable biomass fired improved cookstoves programme for households in Burundi by OBEN E.A.C SA" developed by OBEN E.A.C SA where target population is households.

Aaron Ntakarutimana, Local Councilor, Commune of Kamesa-Musaga, Bujumbura Mairie Province Since the project is of national interest, could you not convince the Government to fund free cookstoves for the benefit of the population?	[answers by Ezeckiel Nibigira, National Deputy, Bujumbura Mairie and Rural Province] The current policy aims at raising people's awareness about changing habits to consume less and be more concerned by environment preservation, based on everyone's means. Government support is ensured in many ways, including facilitation of access to renewable biomass and administrative support, but Government cannot afford financial support.
Agnès Hakizimana, Local Councilor, Commune of Murwi, Cibitoke Province If woodfuel is phased out, how will the less fortunate who couldn't acquire the improved cookstove cope? How do you to commercialize your improved cookstoves and renewable biomass? What is OBEN E.A.C SA commercialization strategy?	The objective is to decrease wood fuel consumption as much as possible but totally suppressing is not possible. Anyway, all efforts are made to bring the stove price affordable to all. OBEN E.A.C SA has taken the challenge up of developing markets for improved cookstoves and renewable biomass in Burundi by: Leveraging mass marketing resources and traditional trade networks. Working with existing local dealer networks that market, sell, and distribute products while pragmatically ramping up depending on penetration rates. Teaming up with NGOs and Microfinance Institutions to take advantage of a pre-existing network of relationships.

Participant's questions/concerns	Project proponent's answers
Innocent Ngomirakiza, Association « Action ceinture verte pour l'environnement (ACVE) » ACVE implements projects to protect the riversides in Bujumbura through afforestation. Our association is 100% behind OBEN E.A.C SA project; We are asking to be associated through afforestation. How can low-income users with no access to banks of microfinance still acquire the stoves? ¹⁴ Can the stoves accommodate for locally made brick pots?	OBEN E.A.C SA sees no inconvenient for partnering with ACVE. Thank you for your interest in submitting a partnership proposal to OBEN E.A.C SA. We will contact you soon. Low-income Burundian households disconnected from retail markets will be able to access the stoves through the local NGOs identified as sales partners; local authorities' network will also be leveraged on. The use of local brick pans is getting very marginal given the availability and affordability of metal pots countrywide.

¹⁴ This question regarded the PoA "Renewable biomass fired improved cookstoves programme for households in Burundi by OBEN E.A.C SA" developed by OBEN E.A.C SA where target population is households.

Oscar Ntirandekura, Neighborhood Leader of Twinyoni, Kamenge Commune Do you plan information and demonstrations on improved cookstoves use in neighborhoods? What woodfuel will be used before the plantations start yielding wood?	The sale outlets will be places of information and demonstrations on stove use. A booklet will also be available for end users. Government has awarded OBEN E.A.C SA with mature forests to sustainably clean and maintain, duty which wasn't performed for the last 20 years. Hence the immediately available renewable wood source.
Participant's questions/concerns	Project proponent's answers
Shemsa Ciza, Local Councillor, Commune of Kinama, Bujumbura Mairie Province Is it possible to grill meat/fish like on a BBQ? Why improved cookstoves are effective in terms of reduction on health impacts?	Yes indeed, you only need to install an appropriate grate. According to the World Health Organization, exposure to smoke from open fires and traditional cookstoves leads to pneumonia, chronic respiratory disease, and lung cancer, etc. With an ICS, you reduce pollution emissions, and you reduce kitchen concentrations of carbon monoxide. You reduce disease and save lives by decreasing exposure to indoor air pollution. Note that fuelwood needs to be well dried, which OBEN E.A.C SA will ensure before commercializing it.

Mechanism for on-going communication with local stakeholders

During the Local Stakeholder consultations, the contacts and addresses of OBEN have been shared with the participants. Participants were encouraged to provide inputs, comments whether by phone, by email or directly in OBEN offices in Bujumbura.

Ongoing communication with restaurant owners is achieved following these two mechanisms:

- 1) During training provided to Restaurants owners, OBEN once again shared with the cookstoves users contacts details and encouraged them to provide inputs if any issues was found during usage of the stoves with briquettes.
- 2) OBEN supply briquettes monthly, the technicians of OBEN have been trained to collect query/comment/grievance when distributing briquettes. Therefore, each month OBEN teams, can get insights from the restaurants owners regarding the stoves or the briquettes. The only issue that was raised was regarding some defects on the grate of the stoves. Each time this issue was raised, OBEN whether repaired/changed the stove directly. The phone number and the address of OBEN are also mentioned in each of the briquettes supplying invoices, ensuring a continuous communication with the restaurant's owners.

2.3 Environmental Impact

No negative environmental impacts have been identified for the project and an environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

As guaranteed to all participants, all recommendations were highly valued and taken into consideration promptly, including:

- Full information availability (on request to project representatives whose coordinates were handed out during consultation, in flyers, newspapers and posters).
- Best efforts to ensure a rapid ramp-up and provide sufficient improved cookstoves and renewable biomass to match the widely enthusiastic demand.
- Continuous communication with community leaders and authorities to fine-tune the approach and cope with any issue for resolution.

2.5 AFOLU-Specific Safeguards

Not Applicable

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The approved baseline and monitoring methodology applied to the project activity is Version 12.0 of AMS-I.E. – “Switch from non-renewable biomass for thermal applications by the user”, available at UNFCCC AMS-I.E.

The CDM tool 30 has also been used for the calculation of the fraction of non-renewable biomass

3.2 Applicability of Methodology

Methodology AMS I.E	Applicability to the project Activity
This methodology comprises of activities to displace the use of non-renewable biomass by introducing renewable energy technologies. 1 Examples of these technologies include, but are not limited to biogas stoves, bio-ethanol stoves, solar cookers, 2 passive solar homes.	Applicable. The project replaces non-renewable biomass used by introducing renewable biomass briquettes to ICS in restaurants
Project participants can show that non-renewable biomass has been used since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	Applicable. It has been proved in section 1.3 of project eligibility

The methodology is applicable for technologies displacing use of non-renewable biomass by renewable energy	Applicable. The project replaces non-renewable biomass used by introducing renewable biomass briquettes to ICS in restaurants
Project participants or coordinating and managing entities shall describe in the PDD/PoA-DD how the double counting of emission reductions has been addressed (e.g., between end users, distributors, and producers of stoves).	PP has defined the unique identification number for each restaurant to avoid double counting
For project activities introducing bioethanol cookstoves, project participants or coordinating and managing entities shall demonstrate that the bioethanol cookstoves are designed, constructed and operated to the requirements (e.g., with regard to safety) of a relevant national or local standard or comparable literature. Latest guidelines issued by a relevant national authority, or an international organization may also be used.	Not applicable, the project does not distribute bioethanol cookstoves
The CDM-PDD or CDM-PoA-DD/CPA-DD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo).	To avoid double counting, the project restaurants the PP has given each restaurant an identification number and PP has given declaration that project will not claim credits in any other programme
The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others also claim credit for emission reductions from the project devices.	Each restaurant participating has signed a declaration to confirm that they will not claim credits from another programme

3.3 Project Boundary

As per the methodology AMS-I.E, the project boundary is the physical, geographical site of the use of biomass. Therefore, it encompasses all the restaurants of Burundi urban centres targeted by the project activity, who will consume the renewable biomass, enabled by the implementation of the improved cookstoves. Only the restaurants using the cookstoves

introduced by the project and the renewable biomass supplied by the project are included in the project boundaries, therefore neither restaurants using project's stove without project's renewable biomass nor restaurants using project's renewable biomass without project's cookstoves will be part of it.

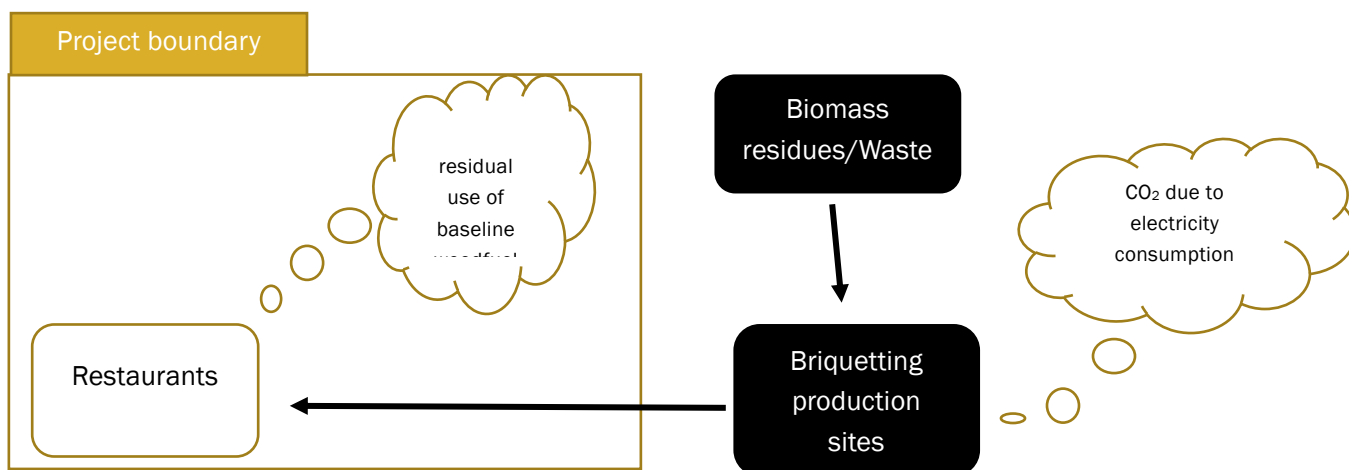
The briquetting machines are not included within the project's boundaries and will rather be accounted as leakage emissions since each production site will supply independently the closest cities where participating restaurants are located but not only because other communities such as schools and police camps and prisons may be supplied as well by the same production sites. Moreover, not any production site will be dedicated specifically to one city's restaurants.

The main emission sources and type of GHGs in the project boundary are CO₂ emissions from substitution of fossil fuel consumption by similar consumers, equivalent to the non-renewable woody biomass saved by the project activity, as listed in the table below:

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Main source of emissions
	CH ₄	No	Minor emissions sources (neglected for simplification)
	N ₂ O	No	Minor emissions sources (neglected for simplification)
	Other	No	Minor emissions sources (neglected for simplification)
Project	CO ₂	Yes	Main source of emissions
	CH ₄	No	Minor emissions sources (neglected for simplification)
	N ₂ O	No	Minor emissions sources (neglected for simplification)
	Other	No	Minor emissions sources (neglected for simplification)
	CO ₂	No	
	CH ₄		

Source	Gas	Included?	Justification/Explanation
	N ₂ O		The renewable biomass used is all from residue/waste, this emission source is not applicable to this project.
	Other		
	CO ₂	No	The renewable biomass used is all from residue/waste, this emission source is not applicable to this project.
	CH ₄		
	N ₂ O		
	Other		
	CO ₂	Yes	Main source of emissions
	CH ₄	No	Minor emissions sources (neglected for simplification)
	N ₂ O	No	Minor emissions sources (neglected for simplification)
	Other	No	Minor emissions sources (neglected for simplification)
	CO ₂	No	Taking into consideration that: the average distance of collection of renewable biomass for the project activity is shorter than the mean distance of origin of previously used biomass in the baseline and; for the proposed project, the residues will never be transported over a distance more than 200 km thus can be neglected. Moreover, the average distance of transportation of renewable briquettes from briquetting plants to restaurants is shorter than the baseline situation where non-renewable woody biomass was transported over a longer distance from isolated rural production site thus can be neglected.
	CO ₂		
	CH ₄		
	Other		
	N ₂ O		
	CO ₂	No	Minor emission source (neglected for simplification).
	CH ₄	No	
	Other	No	
	N ₂ O	No	

Source	Gas	Included?	Justification/Explanation
Leakage	CH ₄	No	Not applicable as the renewable biomass is from waste residues
	CO ₂	No	
	Other	No	
	N ₂ O	No	



3.4 Baseline Scenario

The baseline scenario is the same as the scenario existing prior to the implementation of the project activity, although according to methodology AMS-I.E: It is assumed that in the absence of the project activity, the baseline scenario would be the use of fossil fuels for meeting similar thermal energy needs. Study from university's laboratory CRUEA have highlighted a strong prevailing practice of cooking with traditional low efficiency open stoves supplied with non-renewable wood fuel in most restaurants of Burundi urban centers, resulting in severe deforestation and higher emissions than the proposed CDM project activity. The historical baseline description for technologies and measures has been provided in section 1.10

3.5 Additionality

As per tool 19 "Demonstration of additionality of microscale project activities" Version 09 §11: Page 5: "Project activities that employ renewable energy technology up to 5 MW installed capacity are additional if any one of the conditions below is satisfied":

- (a) The geographic location of the project activity is in one of the least developed countries or the small island developing States (LDCs/SIDS) or in a SUZ of the host country.

- (c) The project activity consists of one or more of the following technologies/measures for distributed energy generation (not connected to a national or regional grid) where end users are households, communities, or small and medium-sized enterprises (SMEs).
- (i) Solar technologies (photovoltaic and solar thermal electricity generation).
 - (ii) Building-integrated wind turbines or rooftop wind turbines.
 - (iii) Micro/pico-hydro.
 - (iv) Micro/pico-wind turbine.
 - (v) PV-Wind hybrid.
 - (vi) Geothermal.
 - (vii) Biomass gasification/biogas.
 - (viii) Solar water heating system.
 - (ix) Clean and energy efficient cookstoves.

The present project is in a Least Development Country and distribute clean cookstoves using renewable energy. The penetration rate for clean fuels and technologies for cooking in Burundi is below 5% (Tracking SDG 7, 2019). Therefore, the conditions are sufficient for the proposed project to be automatically additional

3.1 Methodology Deviations

No methodology deviation has occurred during this monitoring period.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

It is assumed that in the absence of the project activity, the baseline scenario would be the use of fossil fuels for meeting similar thermal energy needs.

Baseline emissions are calculated as:

$$BE_y = B_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel}$$

Where:

$$BE_y = \text{Baseline emissions in the year } y \text{ (tCO}_2\text{e)}$$

By = Quantity of woody biomass that is substituted or displaced in year y (tonnes)

$f_{NRB, y}$ = Fraction of woody biomass used in the absence of the project activity in year y that can be established as non-renewable biomass (fraction or %)

$NCV_{biomass}$ = Net calorific value of the non-renewable woody biomass that is substituted (TJ/tonne)

$EF_{projected_fossil\ fuel}$ = Emission factor of fossil fuels projected to substitute non-renewable woody biomass by similar consumers (tCO₂e/TJ).

Determination of By

By has been calculated after para 29 option C:

Calculated as the product of the number of persons served per institution multiplied by the number of institutions and the estimate of average annual consumption of woody biomass per person that is displaced by the project activity:

$$By = \sum N_{IN,y} \times N_{p,IN,y} \times (BC_{BL,PP} - BC_{PJ,PP,y})$$

Where:

$N_{p,IN,y}$ = Average number of persons served per institution¹⁵ in year y (number)

$N_{IN,y}$ = Number of institutions with functional cookstoves distributed under the project activity in year y (number)

$BC_{BL,PP}$ = Average annual consumption of woody biomass per person before the start of the project activity or at the renewal of each crediting period whichever is later (tonnes/person/year)

$BC_{PJ,PP,y}$ = Average annual consumption of woody biomass per person in the pre project devices during the project activity (tonnes/person/year). This parameter shall be considered if it is found that pre-project devices were not completely displaced but continue to be used to some extent.

The term $(BC_{BL,PP} - BC_{PJ,PP,y})$ refers to the biomass substituted by the project activity.

Determination of Total number of participating restaurant in year y ($N_{IN,y}$)

Baseline survey¹⁶ provides with the number of participating restaurants for the communes of Buterere, Buyenzi, Bwiza, Cibitoke, Gihosha, Kamenge, Kinama, Kinindo, Musaga, Ngagara, Nyakabiga and Rohero displayed in the table below.

The project targets restaurants of all communes of Bujumbura Mairie including Kanyosha and

¹⁵ Institutions refers to restaurants in this project

¹⁶ Staff from OBEN E.A.C SA has collected data from registries of each city hall to calculate the number of registered restaurant per urban center. Excel sheet gathering data is available on ER calculation sheet

secondary urban centers where data are not available at the time of validation therefore they are estimated ex-ante based on the number of restaurants per capita calculated by dividing the number of restaurants identified in the communes included in the baseline survey¹⁷ by global population of these communes in 2008 (République du Burundi, 2008). A value of 0.001636 restaurants per person is used.

BUJUMBURA MAIRIE			
COMMUNES	i	Population 2008 (pers)	Number of participating Restaurants
BUTERERE	1	28,371	75
BUYENZI	2	47,363	103
BWIZA	3	37,688	59
CIBITOKÉ	4	50,899	55
GIHOSHA	5	39,503	80
KAMENGE	6	50,070	89
KANYOSHA	7	59,181	97
KINAMA	8	49,776	45
KININDO	9	21,920	64
MUSAGA	10	43,735	18
NGAGARA	11	30,296	59
NYAKABIGA	12	20,883	35
ROHERO	13	17,481	34
Total		497,166	813
Number of restaurants per capita		$813/497,166 = 0.001635$	Restaurant per person in Urban Centres of Burundi

To estimate the number of participating restaurants in the commune of Kanyosha and the secondary urban centers of Gitega, Ngozi, Kirundo, Kayanza and Rumonge the population of these communes is multiplied by the number of restaurants per capita. Estimated number of participating restaurants is provided in table below.

¹⁷ Communes of Buterere, Buyenzi, Bwiza, Cibitoke, Gihosha, Kamenge, Kinama, Kinindo, Musaga, Ngagara, Nyakabiga and Rohero.

COMMUNES	Population 2008 (pers)	Number of participating Restaurants
KANYOSHA	59,181	253
GITEGA	155,005	103
NGOZI	120,157	196
KIRUNDO	93,110	152
KAYANZA	97,252	159
RUMONGE	145,074	237
TOTAL	610, 598	998

Therefore the total number of restaurants ex-ante from both Bujumbura Mairie and communes are 1,811.

$M_{\text{biomass_substituted},y}$ is determined as follows:

$$M_{\text{biomass_substituted},y} = M_{\text{biomass},y} * (1 - p_{\text{residual_NRB},y})$$

Where:

$M_{\text{biomass_substituted},y}$ = Average annual consumption of substituted woody biomass per appliance in year y (tonnes/year)

$M_{\text{biomass},y}$ = Estimate of average annual consumption of woody biomass per appliance in baseline (tonnes/year)

$p_{\text{residual_NRB},y}$ Proportion of residual use of baseline wood-fuels in year y

The term $M_{\text{biomass},y}$ is determined as follows:

$$M_{\text{biomass},y} = M_{\text{biomass_hist_meal}} * N_{\text{meals_restaurant},y}$$

Where:

$M_{\text{biomass},y}$ = Estimate of average annual consumption of woody biomass per appliance in baseline (tonnes/year)

$M_{\text{biomass_hist_meal}}$ = Average historical consumption of woodfuel equivalent per meal (Tonnes of wood equivalent/meal)

$N_{\text{meals_restaurant},y}$ = Average number of meals prepared per restaurant in the year y (meals/y)

Historical average consumption of woody biomass per meal prepared by the restaurant in the absence of the project activity ($M_{\text{biomass_hist_meal}}$)

Baseline survey provides the following historical data:

The historical wood fuel (in stere ¹⁸/day) and charcoal (in bags ¹⁹/day) consumption of 99 restaurants present in two communes of Bujumbura ²⁰ during survey period in the year 2012.

Conversion of data into woodfuel equivalent:

Conversion from stere of woodfuel to mass of woodfuel equivalent:

A conservative factor of 350 kg/stere is used for converting stere into mass of woodfuel (GTZ-HERA, 2012).

Conversion from bags of charcoal to mass of woodfuel equivalent:

A default value of 6 is applied according to recommendation of the IPCC Guidelines (IPCC 1996- Reference Manual, Chapter 1, Energy, p 1.45) because the production of charcoal is not standardized and suitable local data for estimating the amount of carbon released through charcoal production and consumption are not available. One bag weights 23 Kg²¹ of charcoal.

Conversion from daily basis into yearly basis: Surveys data are provided daily. Data are multiplied by 365 days to convert into yearly basis.

Determination of average historical consumption of woodfuel equivalent per meal distributed by restaurants ($M_{biomass_hist_meal}$)

The average historical consumption of woodfuel equivalent per meal distributed by restaurants in 2012 is determined by dividing the overall historical consumption of woodfuel equivalent of restaurants of communes followed during the baseline survey (Tonnes of woodfuel equivalent per day) divided by the overall number of meals distributed by restaurants of communes followed during baseline survey in 2012 (meals/day).

Table 6: Historical consumption of woodfuel equivalent

	Restaurant's Number having participated to the Baseline survey	Meals number per day	Annual consumption of wood equivalent
Communes	restaurant	Meals/ day	tonnes of wood equivalent/restaurant/y
BUYENZI	43	2,760	69.3
KAMENGE	56	3,930	119.46
Total	99	6,690	97.4

¹⁸ The values are provided in stere per year. 1 stere of wood fuel is equivalent to 1 cubic meter

¹⁹ One bag of charcoal weights 23 kg.

²⁰ Buyenzi and Kamenge

²¹ This is confirmed in the CRUEA's baseline study report

The value of average historical consumption of woodfuel equivalent per meal distributed by restaurants of Burundi is equal to the total historical consumption of all communes surveyed by the total number of meals distributed thus equal to 0.00395 (tonnes of woodfuel equivalent per meal). A discount factor of 0.8922 is applied to account for model uncertainties therefore ($M_{\text{biomass_hist_meal}}$) is taken at $0.00395 * 0.89 = 0.00351$ tonnes of woodfuel equivalent per meal. It corresponds to estimated uncertainty range greater than 30% and less than or equal to 50%.

Average number of meals prepared by restaurant in the year, $y(N_{\text{meals_restaurant},y})$

The average number of meals prepared per restaurant is calculated ex-ante for emissions reduction estimation based on results obtained from a baseline survey available to the DOE. During baseline survey, woodfuel and charcoal consumption of 99 restaurants from two communes of Bujumbura mairie have been followed (Buyenzi and Kamenge) and number of meals distributed per restaurant per day determined. The value of 24,665 meals per year discounted by a factor of 0.89 giving 21,952 is taken based on Baseline survey results for ex-ante estimations.

For ex-post estimation, the average number of meals distributed per appliance in the year y will be monitored based on a sampled number of restaurants n .

A new baseline survey has been conducted in 2020 on 15 project restaurants, to estimate briquettes and number of meals prepared by the restaurants equipped with stoves and briquettes, this survey has been made for 3 days and the results are below:

N°	NAME	LOCALITY	Number of meals per day	Meals per year	Quantity of briquette per meal (kg/meal)
1	Bar ARRET BUS	GITEGA	279	101,713	0,13
2	MARIMBU OUEST HOUSE	RUMONGE	340	124,100	0,13
3	KU MUKENKE	KIRUNDO	255	931,97	0,12
4	RESTAURANT BISMILAH	BUYENZI/MUKAZA	267	97,577	0,09
5	HOTEL LA CASA	GIHOSHA/MUKAZA	475	173,497	0,10
6	RESTAURANT CHEZ MAMAN SOLO	BWIZA/MUKAZA	308	112,542	0,09
7	RESTAURANT POPULAIRE	ROHERO/MUKAA	561	204,643	0,08
8	RESTAURANT AKANOVERA	NYAKABIGA/MUKAZA	412	150,380	0,09
9	RESTAURANT VYISI	KANYOSHA/MUHA	586	213,768	0,07
10	HOTEL ISHIMWE	KININDO/MUHA	614	224,110	0,08
11	SNACK BAR	KININDO/MUHA	442	161,208	0,09

²² The value of 0.89 is used to account for model uncertainties. According to the "Report of the subsidiary body for scientific and technological advice on its eighteenth session, held at Bonn, from 4 to 13 June 2003" Annex III para 6 p25 accessible here: <http://unfccc.int/resource/docs/2003/sbsta/10a02.pdf> . As an example, the value is used in methodology AMS-III.H.

	KUMAZIRAMERO				
12	RESTAURANT HAKORAMUNDA	KINAMA/NTAHANGWA	433	157,923	0,11
13	RESTAURANT KANI	BUTERERE/NTAHANGWA	308	112,542	0,12
14	BAR BIERE LOCALE	NGAGARA/NTAHANWA	270	98,550	0,09
15	RESTAURANT BONNE VIE	CIBITOKI/NTAHANWA	430	156,828	0,09
Average			399	145,505	0,10

Determination of Net calorific value of the substituted non-renewable woody biomass (NCV_{biomass})

According to methodology AMS-I.E, the Net Calorific Value of the non-renewable woody biomass that is substituted has to be taken as IPCC default for wood fuel, 0.0156 TJ/tonne.

Determination of $EF_{projected_fossil\ fuel}$:

The emission factor of fossil fuels projected to substitute non-renewable woody biomass, is determined based on AMS. I.E methodology para 26 table 2 default value. Therefore, the default value for Sub-saharan Africa of 73.2 tCO₂e/TJ have been applied.

Determination of $f_{NRB,y}$

The value of fNRB shall be calculated using either of the following two options:

- Ex ante: the fNRB value is determined once at the validation stage, thus no monitoring and recalculation of the fNRB value during the crediting period is required.
- Ex post: the fNRB_y value is determined for the year y in the crediting period, requiring the fNRB value to be updated annually, following a consistent calculation procedure throughout the crediting period.

The $f_{NRB,y}$ for Burundi have been calculated ex-ante as per option a following CDM tool 30 for multiples project in Burundi and the value is 81.83%. Details of $f_{NRB,y}$ calculation are found in the Emission reduction calculation sheet

4.2 Project Emissions

According to the methodology para 36:

If the project cookstoves use biomass, the following sources of project emissions shall be considered as applicable, bearing in mind that some sources may be only relevant for specific fuels (e.g. production of bioethanol):

Project emissions from cultivation, use and processing of biomass shall be calculated using the latest version of "TOOL16: Project and leakage emissions from biomass" including

- CO₂ emissions from on-site consumption of fossil fuels due to the project activity, (e.g. consumption of fossil fuels for any processing of feedstock);
- CO₂ emissions from electricity consumption by the project activity (e.g. the consumption of electricity for any processing of feedstock);

Project emissions related to cultivation of feedstock.

Project emissions from transportation of biomass if the transportation distance is more than 200 km; otherwise, they can be neglected.

Methane emission from solid waste disposal or wastewater calculated as per provisions in AMS-III.G. (landfill); AMS-III.F. (composting) and AMS-III.H. (wastewater treatment) in cases where the wastes resulting from production and processing of biomass is disposed in anaerobic conditions.

furthermore, as per tool 16. V4 "Project and leakage emissions from biomass". Following project emissions must be considered:

Project emissions from cultivation of biomass:

As the renewable biomass used is all from residue/waste, this emission source is not applicable to this project.

Project emissions from utilization of biomass residues that account for:

Emissions resulting from energy consumption as per para 30,
Emissions resulting from energy consumption are estimated, unless otherwise required in the relevant methodology, by following the provisions in the “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion” and the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”. These emissions include emissions due to thermal and mechanical processing of the biomass residue. In the project activity, there is no leakage or project emissions linked to consumption of fuel. Therefore, the only emissions come from electricity consumption for transformation of renewable biomass into briquettes. These emissions are calculated as follow:

Emissions related to the transformation of renewable biomass into briquettes

Potential project emissions may arise from the electricity consumption from the machines processing the transformation from renewable biomass into briquettes²³. This emissions source is calculated as per the tool ‘Tool to calculate baseline, project and/or leakage emissions from electricity consumption’ version 3. A conservative specific electricity consumption per tonnes of briquettes produced is estimated ex-ante.

Project emissions from electricity consumption

Project emissions from consumption of electricity are calculated, as follows:

$$PE = PE_{EC,y} = \sum EC_{PJ,j,y} * EF_{EF,j,y} * (1 + TDL_{j,y})$$

Where:

$PE_{EC,y}$ = Project emissions from electricity consumption in year y (tCO2/yr)

$EC_{PJ,j,y}$ = Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)

$EF_{EL,j,y}$ = Emission factor for electricity generation for source j in year y (tCO2/MWh)

$TDL_{j,y}$ = Average technical transmission and distribution losses for providing electricity to source j in year y

j = Sources of electricity consumption in the project

Determination of the quantity of electricity consumed by the project ($EC_{PJ,j,y}$)

The renewable briquettes production sites are likely to provide briquettes to different projects therefore it is not possible to monitor electricity consumption for the project alone. Thus the quantity of electricity consumed by the project $EC_{PJ,j,y}$ is calculated based on a default value for the specific quantity of electricity consumed per ton of briquettes produced SEC_{brq} (MWh/ton) estimated ex-ante multiplied by the quantity of briquettes supplied to the project activity $M_{renewable.biomass,y}$.

Ex-ante Determination of the Quantity of renewable biomass consumed by the project activity in year y ($M_{renewable.biomass,y}$)

²³ The briquettes production sites are not included within the boundaries of the project because they will supply different separated projects, different communities and because it is not possible to allocate one production site to one specific city's restaurants as this will depend on the capacity production per site and the regional demand.

For ex-ante estimation of project emissions, the assumption made is:

$$M_{\text{renewable.biomass},y} = \frac{1}{2} * M_{\text{substituted.biomass},y}$$

Value applied ex ante is based on conservative biomass savings estimate derived from preliminary experience²⁴.

Determination of the emission factor for electricity generation ($EF_{EL,j,y}$)

The determination of the emission factor for electricity generation EF depends on scenario applied to the source of electricity consumption:

Scenario A: Electricity consumption from the grid

In this case, project participants may choose among the following options:

- Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the Tool to calculate the emission factor for an electricity system. ($EF_{EL,j/k/l,y} = EF_{\text{grid},CM,y}$).
- Option A2: Use the following conservative default values:
 A value of 1.3 tCO₂/MWh if
 Scenario A applies only to project and/or leakage electricity consumption sources but not to baseline electricity consumption sources; or
 Scenario A applies to: both baseline and project (and/or leakage) electricity consumption sources; and the electricity consumption of the project and leakage sources is greater than the electricity consumption of the baseline sources.
 A value of 0.4 tCO₂/MWh for electricity grids where hydro power plants constitute less than 50% of total grid generation in 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production, and a value of 0.25 tCO₂/MWh for other electricity grids. These values can be used if

Scenario A applies only to baseline electricity consumption sources but not to project or leakage electricity consumption sources; or

Scenario A applies to: both baseline and project (and/or leakage) electricity consumption sources; and the electricity consumption of the baseline sources is greater than the electricity consumption of the project and leakage sources.

Option A2 is chosen conservative default value of 1.3 tCO₂e/MWh.

Average technical transmission and distribution losses for providing electricity to source j in year y ($TDL_{j,y}$)

In case of scenario A, choose one of the following options:

Use recent, accurate and reliable data available within the host country.

Use as default values of 20% for project or leakage electricity consumption sources.

Option chosen default values of 20%.

Small scale project activities may, unless otherwise required by the methodology, neglect emissions from energy consumption associated with processing of biomass residue.

Emissions resulting from transport

Emissions resulting from transport of biomass residue to the processing/utilization facility are estimated, unless otherwise required in the relevant methodology, by following the provisions in the methodological tool "Project and leakage emissions from transportation of freight".

Small scale project activities may, unless otherwise required by the methodology, neglect this source of emissions if the transportation distance is less than 200 km.

²⁴ Experiments to estimate the approximate renewable briquettes consumption per cookstove have been conducted and recorded by project proponent over the same model of cookstoves expected to be used by restaurants in two schools and police camp, evidencing savings up to 90%, thus a conversion factor of 0.5 is a conservative assumption

Transportation of renewable biomass wastes from collection sites to the briquetting plant will be neglected as the briquetting plants will be operated close to the residues provision sites. Moreover, the average distance of transportation of renewable briquettes from briquetting plants to restaurants is shorter than the baseline situation where non-renewable woody biomass was transported over a longer distance from isolated rural production site thus can be neglected.

Moreover, taking into consideration that the average distance of collection of renewable biomass for the Project activity is shorter than the mean distance of origin of previously used biomass in the baseline and;

For the proposed project, the residues/briquettes will never be transported over a distance more than 200 km (one way), then emissions from transportation can be neglected.

Therefore, no additional leakage emissions from transportation need to be accounted for.

4.3 Leakage

The following leakages emissions are considered according to the tool 16 version 4

1) Shifts of pre-project activities

According to tool 16 para 34 “This section is applicable only if the project activity utilizes biomass cultivated in a dedicated plantation. Project proponents are advised to avoid pre-project activities from being shifted outside the project boundary, to avoid indirect land use changes because of the project activity. Rather, project proponents are encouraged to include in the project boundary the land in which the pre-project activities will take place after the project implementation” As the renewable biomass used is all from residue/waste, this emission source is not applicable to this project.

2) Leakage due to diversion of biomass residues from other applications

According to tool 16 para 39, “This section is applicable for project activities which utilize biomass residues. It quantifies leakage due to diversion of biomass residues to the project to be used as either fuel or feedstock. These biomass residues could have been used outside the project boundary in competing applications, and due to the implementation of the project activity these competing application might be forced to use inputs which are not carbon neutral.”. The para 42 guidelines for credible alternative scenario states that: For biomass residues categories for which scenarios B1, B2 or B3 are deemed a plausible alternative scenario, the following procedures should be applied for the combined amount of biomass identified:

- i) Demonstrate that there is an abundant surplus of the biomass residue in the project region which is not utilized. For this purpose, demonstrate that the total quantity of that type of biomass residues annually available in the project region is at least 25 per cent larger than the quantity of biomass residues which is utilized annually in the project region (e.g., for energy generation or as feedstock), including the project facility.
- ii) Demonstrate for the sites from where biomass residues are sourced that the biomass residues have not been collected or utilized (e.g., as fuel, fertilizer, or feedstock) but have been dumped and left to decay, land-filled, left in the field to decay after harvest, or burnt without energy generation (e.g., field burning). This approach is only applicable to biomass residues categories for which project participants can clearly identify the site from where the biomass residues are sourced

The Project activity’s biomass supply involves agro-industrial wastes (sugar cane wastes,

coffee parchments and sawmills residues...) which are widely available in the region and currently burnt without energy purpose²⁵. therefore option B3 para 40 of the tool is applicable. The biomass used in the project activity could not be used for other purposes in the absence of the project.

As per the public report “Elaboration of a strategy for Energy sector in Burundi” issued by Ministry of Energy and Mines, General Directorate of Energy and Water (Republic of Burundi, Ministry of Energy, Water department, 2011), the annual available amount of renewable biomass, which is not used, is about 1,378,374.43 tonnes (including 1,127,302 tonnes of crop residues, 224,966.2 tonnes of forest litter and 26,106.23 tonnes of timber residues). Based on production capacity of 3 briquetting machines of 500 tonnes per month the quantity of biomass needed is 18,000 tonnes of dry biomass, thus 36,000 tonnes of wet biomass residues assuming 50% moisture of biomass residues. Therefore, the quantity of available biomass residues in the region is at least 25% larger than the quantity of biomass that is utilized including the project activity. This source of leakage can thus be neglected.

Methodology AMS-I.E further indicates that leakage related to the non-renewable woody biomass saved by the project activity shall be assessed based on ex post surveys of users and the areas from which this woody biomass is sourced (using 90/30 precision for a selection of samples). The following potential source of leakage shall be considered:

- a) The use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass used by the non-project households/users that is attributable to the project activity, then B_y is adjusted to account for the quantified leakage.

Alternatively, B_y is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required. Therefore, as per the methodology B_y will be adjusted by a factor of 0.95.)

4.4 Net GHG Emission Reductions and Removals

Parameter	Value	Unit	Description of parameter	Source/reference/justifications
Mbiomass_hist_meal	0.00351	Tonnes of wood equivalent / meal	Average historical consumption of woodfuel equivalent per meal (Tonnes of wood equivalent/meal)	Derived from baseline survey discounted by a factor of 0.89.
Nmeals_restaurant,y	145,505	meals/restaurant/year	Average number of meals prepared per restaurant in the year y	Derived from baseline survey.
Mbiomass,y	511.2	Tonnes of Wood equivalent / appliance/y	Estimate of average annual consumption of woody biomass per appliance in baseline	$M_{biomass,y} = M_{biomass_hist_meal} * N_{meals_restaurant,y}$

²⁵ Apart from the cogeneration project of SOSUMO where already residues is declared as too many. Indeed, SOSUMO has too much residues and has to burn a great part of it. That's why, OBEN E.A.C SA is signing a partnership with SOSUMO at the sugar cane plant location in order to process the surplus into briquettes

Nrestaurant,y	855	Restaurant	Total number of participating restaurant in year y	Project proponent expectation.
pop restaurant,y	90	%	Proportion of restaurants still operating ICS or having replaced their ICS by an equivalent in service one under similar conditions under the project activity in year y.	Based on project proponent expectations and experience.
Noperating_restaurant,y	769	unit	Number of restaurants still operating or where ICS has been replaced by an equivalent in service appliance under the project activity in year y	Noperating_restaurant,y = Nrestaurant,y * pop restaurant,y
Presidual_NRB,y	10	%	Proportion of residual use of baseline wood-fuels in year y	The proportion of residual use of woody biomass is estimated ex- ante at 10%, based on project planner's knowledge and experience.
Mbiomass_substituted,y	460.08	tonnes/ y	Average annual consumption of substituted woody biomass per appliance (tonnes/year)	Mbiomass_substituted,y = Mbiomass,y * (1 - Presidual_NRB,y)
Leakage _{adj}	0.95	-	Net to gross adjustment factor to account for leakages	Default Value
B _y	437	tonnes/y	Quantity of woody biomass that is substituted or displaced in tonnes	B _y = Leakage _{adj} * Mbiomass_substituted,y * Noperating_restaurant,y
f _{NRB,y}	81,83	%	Fraction of biomass used in the absence of the CDM project in year y that can be established as non-	Calculated
NCV _{biomass}	0.015	TJ/ton	Net calorific value of the non-renewable woody biomass that is substituted	Default Value

EF _{projected_fossilfuel}	73,2	tCO ₂ /TJ	Emission factor for the substitution of non-renewable woody biomass by similar consumers	Default Value
SEC _{briq}	0.038	MWh/Metric tonne	Default value for the specific quantity of electricity consumed per ton of briquettes produced	Default Value provided by technology provider
M _{renewable.biomass,y}	11,079	tonnes/y	Quantity of renewable biomass consumed by the project activity in year y.	The parameter will be used for monitoring project emissions. For ex-ante estimation the assumption made is: M _{renewable.biomass,y} = 1/2 * M _{substituted.biomass,y}
ECP _{j,y}	8	MWh/y	Quantity of electricity consumed by the project electricity consumption source j in year y	ECP _{j,y} = SEC _{briq,y} * M _{renewable.biomass,y}
EF _{EL,j,y}	1.30	tCO ₂ /MWh	Emission factor for electricity generation for source j in year y (tCO ₂ /MWh)	Default Value
TDL _{j,y}	0.20	-	Average technical transmission and distribution losses for providing electricity to source j in year y	Default Value
LEEC _y	13	tCO ₂ /yr	Project emissions from electricity consumption in year y	Calculated

From the parameters above, the emissions reduction ex-ante are:

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})

03/02/2020 to 31/12/2020	226 164	107 051	-	119 112
01/01/2021 to 31/12/2021	303 215	143 522	-	159 692
01/01/2022 to 31/12/2022	303 215	143 522	-	159 692
01/01/2023 to 31/12/2023	303 215	143 522	-	159 692
01/01/2024 to 31/12/2024	303 215	143 522	-	159 692
01/01/2025 to 31/12/2025	303 215	143 522	-	159 692
01/01/2026 to 31/12/2026	303 215	143 522	-	159 692
01/01/2027 to 31/01/2027	25 268	11 960	-	13 307
Total	2,070,721	824,663	-	917,572
Total number of crediting years	7			
Annual average over the crediting period	295,817		117,809	131,082

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	P_j			
Data unit	kW _{th}			
Description	Useful thermal output capacity of the ICS type j			
Source of data	Performance test results			
Value applied:	ICS type j	Medium	Large	Matawi
	Useful thermal output capacity	2.49 kW _{th}	4.82 kW _{th}	1.45 kW _{th}
Justification of choice of data or description of measurement methods and procedures applied	Useful thermal output of the ICS has been estimated by Water Boiling Test by the Turkish supplier for the Turkish stoves and by the CRUEA laboratory for OBEN Matawi stoves			
Purpose of Data	Small-scale threshold, debundle and automatic additionality compliance			
Comments	-			

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of biomass used in the absence of the CDM project in year y that can be established as non-renewable biomass
Source of data	Forest resources assessments 2015 (FAO)
Value applied:	0.8183
Justification of choice of data or description of measurement methods and procedures applied	The fraction of non-renewable biomass has been calculated following the CDM tool 30

Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	NCV _{biomass}
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied:	0.015
Justification of choice of data or description of measurement methods and procedures applied	According to methodology AMS-I.E, the Net Calorific Value of the non-renewable woody biomass that is substituted has to be taken as IPCC default for wood fuel.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{projected_fossilfuel}
Data unit	tCO ₂ /TJ
Description	Emission factor for the substitution of non-renewable woody biomass by similar consumers
Source of data	AMS-I.E version 12.0
Value applied:	73.2
Justification of choice of data or description of measurement methods and procedures applied	As indicated by the methodology AMS-I.E, this value represents the emission factor of the substitution fuels likely to be used by similar users in Sub-Saharan Africa
Purpose of Data	Calculation of baseline emissions

Comments	-
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Data / Parameter	Mbiomass_hist_meal
Data unit	Tonnes of wood equivalent / meal
Description	Average historical consumption of woodfuel equivalent per meal (Tonnesof wood equivalent/meal)
Source of data	Derived from baseline survey discounted by a factor of 0.89.
Value applied:	0.00351
Justification of choice of data or description of measurement methods and procedures applied	As permitted by the methodology AMS-I.E, this was derived from Baseline survey gathering restaurants woodfuel consumption data of two communes of Bujumbura. The historical consumption of restaurants has been divided by the historical number of meals distributed in order to get the historical consumption of woodfuel equivalent per meal. The value obtained has been discounted by a factor of 0.89 to account for model uncertainties.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	Leakage _{adj}
Data unit	fraction
Description	Net to gross adjustment factor to account for leakages
Source of data	Methodology AMS-I.E Version 12
Value applied:	0.95
Justification of choice of data or description of measurement methods and procedures applied	In case this leakage adjustment factor is applied, it is not required to survey
Purpose of Data	Calculation of leakage emissions

Comments	the use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources.
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Data / Parameter	EFEL,j,y
Data unit	tCO ₂ /MWh
Description	Emission factor for electricity generation for source j in year y
Source of data	“Tool to calculate baseline, project and/or leakage emissions from electricity consumption” (Version 03)
Value applied:	1.3
Justification of choice of data or description of measurement methods and procedures applied	Option A2: Use a conservative default value of 1.3 tCO ₂ /MWh
Purpose of Data	Calculation of leakage emissions
Comments	-

Data / Parameter	TDL _{j,y}
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to the briquetting machine
Source of data	“Tool to calculate baseline, project and/or leakage emissions from electricity consumption” (Version 03)
Value applied:	0.20
Justification of choice of data or description of measurement methods and procedures applied	Use as default values of 20% for option: (a) project or leakage electricity consumption sources;

Purpose of Data	Calculation of leakage emissions
Comments	-

Data / Parameter	SEC _{briq}
Data unit	MWh/tonne
Description	Default value for the specific quantity of electricity consumed per tonne of briquettes produced
Source of data	Estimated ex-ante based on historic specific electricity consumption per tonne of briquettes produced.
Value applied:	0.038
Justification of choice of data or description of measurement methods and procedures applied	The specific electricity consumption is estimated ex-ante based on historic data of electricity consumption per tonne of briquettes produced
Purpose of Data	Calculation of leakage emissions
Comments	Specific consumption provided by the constructor is 35-38 kWh/Metric tonne of Biomass. To be conservative, a value of 38 kWh/tonne is chosen.

Data / Parameter	m _{biomass} / m _{charcoal}
Data unit	kg biomass/kg charcoal
Description	Conversion factor wood/charcoal
Source of data	IPCC 1996- Reference Manual, Chapter 1, Energy, p 1.45 Or national/local values from literature, surveys, or expert judgments if available
Value applied:	6
Justification of choice of data or description of	The default value should be applied according to recommendations of the IPCC Guidelines if no national/local

measurement methods and procedures applied	value is available. However, such default value may be revised ex-post would reliable and country-specific data become available during monitoring.
Purpose of Data	Calculation of baseline
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	Ndistributed_ICS_{j,y}
Data unit	Number
Description	Total number of distributed ICS (medium and large) or replaced by an equivalent appliance in year y
Source of data	Project implementer database
Description of measurement methods and procedures applied	Each participating restaurant who will acquire an ICS will have to sign a product acquisition agreement with the project implementer. The project implementer will keep a database of all distributed ICS.
Frequency of monitoring/recording	Continuously
Value applied:	5,233
Monitoring equipment	<i>Not applicable. The number of ICS distributed is calculated by the addition of all the ICS distributed since the start of the project</i>
QA/QC procedures applied	The total number of distributed ICS mentioned in the data base will be cross-check by the project implementer with number of contracts signed with participating restaurants. Databases will be periodically checked by the project implementer to count the total number of distributed ICS under the project.

Purpose of data	Used to record the number/location of each stove distributed and verify the compliance with small scale limit.
Calculation method	PP has an internal control of the quantity of stoves distributed. The number is rechecked according to sales agreement.
Comments	Capped in compliance with the small-scale limit with regard to thermal energy power output of the stoves as detailed in B.2. $N_{distributed_ICS,medium} * P_{medium} + N_{distributed_ICS,large} * P_{large} < 45 \text{ MWth}$ Distribution between medium and large and number of improvedcookstoves is not known at the time of project validation.

Data / Parameter	Nrestaurant,y
Data unit	Number
Description	Total number of participating restaurants in year y
Source of data	Project implementer database
Description of measurement methods and procedures applied	Each participating restaurant who will acquire an ICS will have to sign a product acquisition agreement with the project implementer. The project implementer will keep a database of all distributed ICS and participating restaurant.
Frequency of monitoring/recording	Continuously
Value applied:	852 restaurants
Monitoring equipment	The Number of restaurants is extract from the project database maintained by the PP
QA/QC procedures applied	The total number of restaurants mentioned in the data base will be cross- check by the project implementer with number of contracts signed with participating restaurants. Databases will be periodically checked by the project implementer to count the total number of restaurant under the project..

Purpose of data	Used to record the number/location of each restaurant.
Calculation method	Not applicable
Comments	-

Data / Parameter	Pop restaurant,y
Data unit	fraction
Description	Proportion of restaurants still operating ICS or having replaced their ICS by an equivalent in service one under similar conditions under the project activity in year y.
Source of data	Monitored sample observations.
Description of measurement methods and procedures applied	A physical check will be performed by the CDM project proponent at least annually in accordance with the monitoring plan. According to §12 (a) of the latest Standard for Sampling and Surveys the Proportion of restaurants still operating ICS or having replaced their ICS by an equivalent in service one under similar conditions under the project activity is conservatively estimated ex-ante at 90% for the first monitoring session to be revised during monitoring, based on project planner's knowledge and experience.
Frequency of monitoring/recording	At least annually
Value applied:	0.90
Monitoring equipment	Not applicable
QA/QC procedures applied	$0 \leq \text{Pop restaurant,y} \leq 1$ Database will be periodically checked by the project proponent. Each stove will be traced from manufacturer to user (restaurant) using its unique ID. The count, prorated from the monitoring sample size, cannot exceed the total distributed number of stoves at any point in time and number of participating restaurants, as recorded in the stove sales register. Any invalid or

	incomplete entries related to cook stove sales will not be taken into consideration in Emissions Reductions calculation.
Purpose of data	Calculation of baseline emissions
Calculation method	The parameter is determined ex-ante. However, for verifications, the parameter will be calculated ex-post by determining the number of restaurants no more using the ICS, Suring monitoring surveys
Comments	For subsequent monitoring sessions $p_{\text{op restaurant},y-1}$ measured during the previous monitoring session shall be used as an estimation of $p_{\text{op restaurant},y}$ for the new sample size calculation. Guidance for monitoring is provided to field agents in order to determine the operating status of each appliance during monitoring session with specific conditions to comply with.

Data / Parameter	Presidual_NRB,y
Data unit	fraction
Description	Proportion of residual use of baseline wood-fuels in year y
Source of data	Periodic monitoring visit findings
Description of measurement methods and procedures applied	As per the agreement between end-user and the project implementer, each participating restaurant agrees to record the number of meals cooked with the ICS with renewable biomass (briquettes) and baseline wood-fuels during the monitored period p if or when the restaurant is included in the sample n . During the annual monitoring visit, field agents will establish the quantity of meals cooked with biomass from non-renewable sources. The proportion of residual use of woody biomass from non-renewable sources was estimated ex-ante at 10%, based on project planner's knowledge and experience.
Frequency of monitoring/recording	At least annually
Value applied:	0.10

Monitoring equipment	Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.
QA/QC procedures applied	$0 \leq \text{Presidual_NRB},y \leq 1$ Database will be periodically checked by the project proponent. Abnormal levels of residual usage of baseline wood-fuels will be inquired and documented. Any invalid or incomplete entries related to residual consumption of baseline wood-fuels will dismiss the sampled restaurant from consideration in Emissions Reductions calculation.
Purpose of data	Calculation of baseline emissions
Calculation method	The parameter is determined ex-ante. However, for verifications, the parameter will be calculated ex-post by determining the number of restaurants no more using the ICS, Suring monitoring surveys
Comments	For subsequent monitoring sessions $p_{\text{Presidual_NRB},y-1}$ measured during the previous monitoring session shall be used as an estimation of $p_{\text{Presidual_NRB},y}$ for the new sample size calculation. All eventual remaining biomass consumed from outside of the OBEN E.A.C SA's renewable supply chain is conservatively considered as baseline-wood-fuels.

Data / Parameter	Mrenewable_biomass,y
Data unit	tonnes/year
Description	Quantity of renewable biomass consumed by the participating restaurants in year y
Source of data	Delivery notes from renewable delivery site(s)
Description of measurement methods and procedures applied	Value applied is based on estimations from 2020 delivery notes.
Frequency of monitoring/recording	Every month

Value applied:	11,079
Monitoring equipment	Not applicable
QA/QC procedures applied	The value of $M_{\text{biomass},y}$ multiplied by the number of operating stoves should be a qualitative comparison of consumed renewable and baseline woodfuel with historical consumption (efficiency-adjusted) on a calorific basis.
Purpose of data	Project emissions and Methodology requirements
Calculation method	The parameter will be used for monitoring project emissions from electricity consumption. For ex-ante estimation of project emissions, the assumption made is: $M_{\text{renewable.biomass},y} = 1/2 * M_{\text{substituted.biomass},y}$
Comments	-

Data / Parameter	$N_{\text{meals_restaurant},y}$
Data unit	meals/restaurant/year
Description	Average number of meals prepared per restaurant in the year y (meals/y)
Source of data	Derived from baseline survey conducted
Description of measurement methods and procedures applied	Number of meals prepared per restaurant sampled will be recorded during each monitoring period.
Frequency of monitoring/recording	At least annually
Value applied:	145,505
Monitoring equipment	Not applicable
QA/QC procedures applied	-

Purpose of data	Calculation of baseline emissions
Calculation method	For ex-ante calculation the estimation is based on the data obtained from a baseline survey compiling data from 15 restaurants of Bujumbura. Ex-post calculation will be based on surveys results.
Comments	-

5.3 Monitoring Plan

The monitoring plan defines the procedures to measure and record parameters necessary for the verification of the emission reductions effectively achieved. The monitoring will be carried out according to AMS-I.E. The project is managed by OBEN E.A.C SA who is the project proponent. OBEN E.A.C SA ensures the overall briquetting plant operation safety and cookstove distribution in accordance with Burundi's Laws and technology providers' guidelines. Cookstove distribution's responsibility may be shared with local microfinance institutions able to provide with a credit the ICS to restaurants and be reimbursed over a period of 1 to 2 years thanks to the economy made one the fuel used.

APPENDIX 1: BRIQUETTES FACILITIES

The project technology employs direct compaction (binder-less) equipment – Jumbo-90 Briquetting Machine, most popular in India. Jumbo-90 is a flagship model of briquetting industry with high production, low conversion cost and easy operation. These binders-less briquetting machines are based on very high compaction characteristics of combustible cellulosic agro waste such as bagasse saw dust, groundnut shells, etc. into cylindrical briquettes under high pressure & process heat.

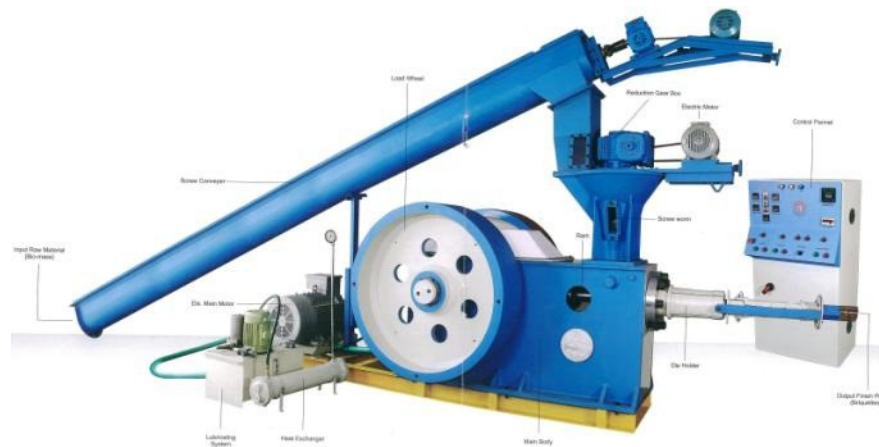


Figure 8 : Jumbo-90 briquetting machine overview

The features of Jumbo-90 are:

- Acceptability up to 20 mm size of raw material. (No need of powdery form).
- High density of finished product with 90 mm diameter.
- Low electric consumption due to direct feeding without hammer mill.
- No loss of production & Air pollution due to direct feeding system.
- Expected lifetime of 25 years.

Table 7: Electricity consumption specifications

	Supplied Elec.Motor (HP) ⁵⁰	Required Load (HP)
Main elec. Motor	75	60
Feeding Kuppy gear	7.5	5
Screw Conveyor	3	2
Lubrication	3	2
Total HP	88.5	69
Total equivalent kW	66kW	35 - 38 kWh/Metric Tonne of biomass

PROCEDURE TO DETERMINE THE USEFUL THERMAL OUTPUT OF ICS

As per the Clean Development Mechanism Project Standard (Version 5.0) to determine the performance of equipment used in the proposed small-scale CDM project activity, project participants shall use:

- (a) The appropriate value specified in the selected methodology.
- (b) The national standard for the performance of the equipment type (project participants shall identify the standard used) if the value specified in subparagraph (a) is not available.
- (c) An international standard for the performance of the equipment type, such as International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC) standards (project participants shall identify the standard used) if the value specified in subparagraph (b) is not available.
- (d) The manufacturer's specifications, provided that they are tested and certified by national or international certifiers, if the value specified in subparagraph (c) is not available.
- (e) Performance data from test results conducted by an independent entity for equipment installed under the project activity if the value specified in subparagraph (d) is not available.

Not any value is specified in methodology (option a is not available), no any national standard for the performance of equipment type is identified (option b is not available) therefore option (c) is used: an international standard is used for determining equipment performance. Water Boiling Test protocol has been used and test results provided to the VVB.

Useful thermal output

The Berkeley's WBT (Water Boiling Test) protocol was used for testing performance. The useful thermal output of a cook stove can be determined as the mean effective thermal power, i.e. the quotient of effective energy delivered for the cooking process divided by heating time. In other words, this corresponds to the average rate of energy released from fuel combustion that is transferred to the pot over the duration of a certified water boiling test.

APPENDIX 2: BIBLIOGRAPHY

1. Cirad. (2017) Utilisation du charbon de bois comme principale source d'énergie de la population urbaine : un sérieux problème pour la conservation du couvert forestier au Burundi
Retrieved from <https://revues.cirad.fr/index.php/BFT/article/download/ID-BFT-151127/31022>
2. EUEI. (2011). Energy Strategy and Action Plan for Burundi. Eschborn, Germany: European Union Energy Initiative.
Retrieved at : https://www.climate-laws.org/rails/active_storage/blobs/eyJfcmFpbHMiOnsibWVzc2FnZSI6IkJBaHBBZ0IHIiwZXBwIjpuZDVsX2IjOiJibG9iX2IklIn19-876efb433cc8c194380a48bf261926ddebd719a8/f
3. GTZ-HERA. (2012). *Manual for Programs and Projects to Implement Cooking Energy Interventions*. Retrieved from https://energypedia.info/wiki/Assessing_Wood_Fuel_Supply_Potentials
4. The Energy Progress Report (tracking SDG 7, 2019) retrieved from https://sustainabledevelopment.un.org/content/documents/2019_Tracking_SDG7_Report.pdf
5. Ministry of Energy, Water department. (2011). *Elaboration of a strategy for Energy sector in Burundi*. (Strategic plan for energy sector in Burundi)
Retrieved from : https://rise.esmap.org/data/files/library/burundi/Documents/Clean%20Cooking/Burundi_%C3%89laboration%20de%20la%20Strat%C3%A9gie%20sectorielle%20pour%20le%20secteur%20de%20l'E2%80%99%C3%A9nergie%20au%20Burundi.pdf
6. République du Burundi. (2008). *Troisième recensement général de la population et de l'habitat*.
Retrieved at : <https://catalog.ihnsn.org/catalog/4167/download/55517>
7. World Health Organisation (2021). *Household air pollution and health*.
Retrieved from : <https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health>
8. World Data Bank. (2020). *Change in forest cover in Burundi between 1990 and 2020*
Retrieved from: <https://data.worldbank.org/indicator/AG.LND.FRST.ZS?locations=BI>